

Data visualisation with clcharts: : CHEAT SHEET

Basics

clcharts creates charts in the House of Commons Library style and is based on [ggplot2](#).

ggplot2 is based on the grammar of graphics, the idea that you can build every graph from the same components: a **data** set, a **coordinate** system, and **geoms**—visual marks that represent data points.

To create a chart we use:

ggplot (data = <DATA>) +

<GEOM FUNCTION>(
 mapping = aes(<MAPPINGS>),
 stat = <STAT>,
 position = <POSITION>
) +

<COORDINATE FUNCTION> +

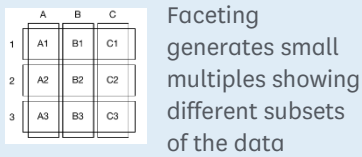
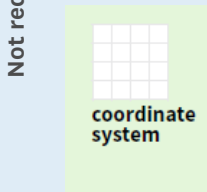
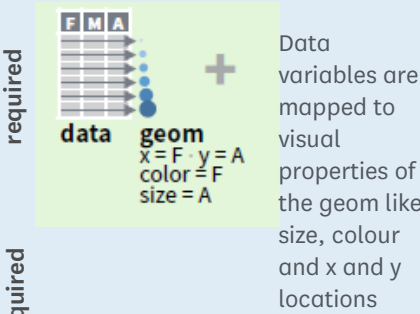
<FACET FUNCTION> +

<SCALE FUNCTION> +

theme_commonslib() +

labs(
 x = <x label>,
 y = <y label>
) +

add_commonslib_title(
 title = <title>,
 subtitle = <subtitle>
)



House of Commons library theme

Set labels e.g. for x and y axes
Set labels e.g. for x and y axes

This aligns the title and subtitle to the left hand edge of the y axis which is consistent with the Library style.

Quickstart

LINE CHART

```
plot <- ggplot(  
  data = <data>,  
  mapping = aes(x = <x>, y = <y>)  
) +
```

```
geom_line(color = commonslib_color("commons_green"), size =  
1.1) +
```

```
labs(  
  x = <x>,  
  y = <y>) +
```

```
scale_x_continuous(breaks = seq(0, 10, 2)) +  
scale_y_continuous(limits = c(0, 20)) +
```

```
theme_commonslib()
```

```
plot <- plot + add_commonslib_titles(  
  plot,  
  title = <title>,  
  subtitle = <title>  
)
```

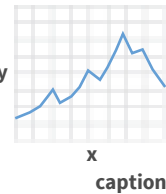
```
# Save the plot  
save_png(plot, "<plotname>.png", width = 8, height = 5)
```

aes maps <x> to x and <y> to y. This can be set here to apply to all geoms, or set in each geom.

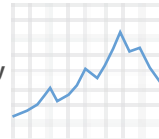
Set x and y axes labels here

Control axes: x is from 0 to 10 in steps of 2, y is between 0 and 20

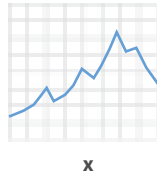
Chart adjustment and annotation



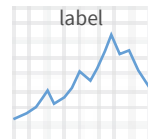
```
labs(caption=<caption>)  
+  
theme_commonslib(  
  caption_position=<left|  
right>  
)
```



```
labs(x=NULL)
```



```
labs(y=NULL)
```



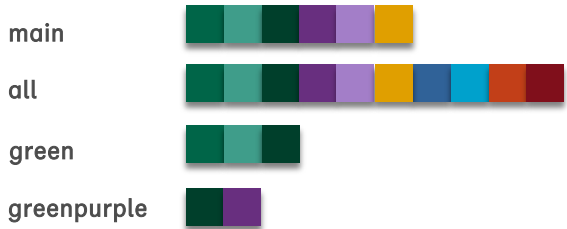
```
annotate_commonslib(  
  x=<x>, y=<y>,  
  label=<label>,  
  size=4.2)  
  
Equivalent to 12pt, size  
is in mm (1 pt  
=0.35mm)
```

Theme colours

commons_green	tangerine
ocean_green	cerulean_blue
pine_green	pacific_blue
grape	burnt_orange
lilac	cherry

commonslib_color(<color_name>)

Palettes



scale_color_commonslib(palette=<palette_name>)

scale_fill_commonslib(palette=<palette_name>)

Saving

```
save_svg(plot, filename, width, height)  
save_png(plot, filename, width, height, dpi)
```

width, height in inches.
dpi = dots per inch (default 400)
100-150dpi, lower resolution
300 dpi: standard
600 dpi: high detail